INCH-POUND
MIL-PRF-24154A(SH)
w/AMENDMENT 3
9 November 2021
SUPERSEDING
MIL-S-24154A(SHIPS)
10 March 1967

PERFORMANCE SPECIFICATION

SYNTACTIC BUOYANCY MATERIAL FOR HIGH HYDROSTATIC PRESSURES

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for rigid syntactic buoyancy materials used to provide buoyancy for deep ocean submerged research vehicles, and other related applications.

1.2 Classification. The syntactic buoyancy material is of the following types, as specified (see 6.2).

a. Type I – For 4,500 pounds per square inch gauge (psig) hydrostatic pressure.

b. Type II – For 10,000 psig hydrostatic pressure.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

DELETED

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY

ASQ Z1.9 - Sampling Procedures and Tables for Inspection by Variable for Percent Nonconforming

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D638 - Standard Test Method for Tensile Properties of Plastics

ASTM D695 - Standard Test Method for Compressive Properties of Rigid Plastics

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 9330

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:42Z
Check the source to verify that this is the current version before use.

ASTM D732 - Standard Test Method for Shear Strength of Plastics by Punch Tool

ASTM D1622 - Standard Test Method for Apparent Density of Rigid Cellular Plastics

ASTM DS56L-EB - Metals and Alloys in the Unified Numbering System (UNS)

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL

SAE J1086 - Numbering Metals and Alloys

(Copies of this document are available online at www.sae.org.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.6) in accordance with 4.3.

3.1.1 DELETED.

3.2 Physical properties. Physical properties of the syntactic buoyancy material shall conform to tables I and II before and after exposure to the appropriate conditioning. Table I is exposed to conditions in 4.6.7.1 and table II is exposed per 4.6.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:42Z
Check the source to verify that this is the current version before use.

TABLE I. Physical requirements before and after 1,000 cycles hydrostatic exposure (first article tests only).

Property [1/]	Type I		Type II	
	Before Exposure	After Exposure [2/]	Before Exposure	After Exposure [3/]
Density (max.) (block) (lbs/cu.ft.) module with skin	38	--	44	--
Density (max.) (lbs/cu.ft.) specimen	38	--	44	--
Water absorption (max.) (percent by weight)	--	3	--	3
Uniaxial ultimate compressive strength lbs/sq.in. (min.)	7,000	6,000	12,500	11,500
0.2 percent offset uniaxial compressive stress, lbs./sq.in. (min.)	5,000	4,250	10,000	9,000
Compressive modulus lbs./sq.in. (min.)	300,000	250,000	425,000	400,000
Tensile strength lbs./sq.in. (min.)	1,500	--	4,000	--
Shear strength lbs./sq.in. (min.)	2,000	--	4,000	--
Bulk modulus lbs./sq.in. (min.)	275,000	--	350,000	--
Hydrostatic compressive strength, lbs./sq.in. (min.)	9,000	--	16,500	--
Hydrostatic compressive strength, max. reduction from initial before exposure strength, percent	--	10	--	10

NOTES:

1/ Density, initial uniaxial compressive strength, initial 0.2 percent offset uniaxial compressive stress, and initial compressive modulus can be determined from one set of specimens; also, water absorption, after exposure uniaxial compressive strength, after exposure 0.2 percent offset uniaxial compressive stress, and after exposure compressive modulus can be determined from another set of specimens.

2/ After 1,000 cycles of hydrostatic exposure between 0 to 4,500 psig.

3/ After 1,000 cycles of hydrostatic exposure between 0 to 10,000 psig.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:42Z
Check the source to verify that this is the current version before use.

TABLE II. Physical requirements before and after one week of static exposure (first article and conformance tests).

Property [1/]	Type I		Type II	
	Before Exposure	After Exposure [2/]	Before Exposure	After Exposure [3/]
Density (max.) (block) (lbs/cu.ft.) module with skin	38	--	44	--
Density (max.) (lbs/cu.ft.) specimen	38	--	44	--
Water absorption (max.) (percent by weight)	--	2	--	2
Uniaxial ultimate compressive strength lbs/sq.in. (min.)	7,000	6,300	12,500	12,000
0.2 percent offset uniaxial compressive stress, lbs./sq.in.(min.)	5,000	4,500	10,000	9,500
Compressive modulus lbs./sq.in. (min.)	300,000	270,000	425,000	400,000
Tensile strength lbs./sq.in. (min.)	1,500	--	4,000	--
Shear strength lbs./sq.in. (min.)	2,000	--	4,000	--
Bulk modulus lbs./sq.in. (min.)	275,000	--	350,000	--
Hydrostatic compressive strength, lbs./sq.in. (min.)	9,000	--	16,500	--
Hydrostatic compressive strength, max. reduction from initial before exposure strength, percent	--	10	--	10

NOTES:

1/ Density, initial uniaxial compressive strength, initial 0.2 percent offset uniaxial compressive stress, and initial compressive modulus can be determined from one set of specimens; also, water absorption, after exposure uniaxial compressive strength, after exposure 0.2 percent offset uniaxial compressive stress, and after exposure compressive modulus can be determined from another set of specimens.

2/ One week (168 hours) exposure at 4,500 psig hydrostatic pressure.

3/ One week (168 hours) exposure at 10,000 psig hydrostatic pressure.

3.3 Material. The syntactic buoyancy material (see 6.3) shall consist of a low-density filler, such as hollow glass microspheres, incorporated in a resin matrix, such as epoxy resin.

3.4 Sizes and forms. The syntactic buoyancy material shall be furnished in modules, blocks, sheet, or other forms, as specified (see 6.2).

3.5 Dimensional tolerances. The dimensions and dimensional tolerances shall be as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:42Z
Check the source to verify that this is the current version before use.

3.6 Color. The color shall be the natural color of the product.

3.7 Resin skin. Syntactic buoyancy material shall be supplied in the shapes and sizes required with or without resin skin or with a specific number of sides with or without resin skin, as specified (see 6.2). The resin skin normally results from the production process. In the event that syntactic buoyance material with resin skin is specified, and the natural skin is damaged, it shall be replaced using the same resin system as the cured foam casting or a resin system approved by the command or agency concerned.

3.8 Workmanship. The syntactic buoyancy material shall be uniform in quality and be free of cracks, splits, seams, porous, or resin-rich areas.

4. VERIFICATION

4.1 Responsibility for inspection. DELETED.

4.2 First article tests. First article tests shall be conducted at a laboratory acceptable to the Technical Authority. First article tests shall consist of the tests specified in 4.6 and shall meet all the requirements of tables I and II.

4.2.1 First article sample. A one-piece casting (see figure 1) of syntactic buoyancy material measuring 12 by 12 by 12 inches plus or minus ¼ inch with a resin skin shall be submitted for first article inspection. The first article sample shall be a single homogenous block and shall not be composed of smaller modules adhered together.

4.3 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination specified in 4.5.2.

4.4 Sampling.

4.4.1 Sampling for first article inspection. Nine each, initial and exposure, compressive, water absorption, shear, and density specimens shall be selected from the 12 by 12 by 12-inch first article sample. Two 1-inch diameter by 2-inch long specimens shall be randomly selected from each of the following nine stratified sections: top-center, top-edge, top-corner, middle-center, middle-edge, middle-corner, bottom-center, bottom-edge, and bottom-corner; figure 1 graphically shows stratification. This provides nine specimens for determining density and initial compressive properties and nine for providing water absorption and after exposure compressive properties, representing each of the non-stratified sections. Five shear specimens shall be randomly selected from the top-corner specimen, middle-corner, bottom-corner, middle-edge, and middle-center. Five tensile, three bulk modulus, and three hydrostatic compressive strength specimens shall be randomly selected from the remainder 12 by 12 by 4-inch section shown on figure 1.

4.4.2 Sampling for conformance inspection. All modules shall be inspected for conformance with 4.5.2.

4.4.3 Sampling for conformance testing. From each lot, modules shall be selected in accordance with ASQ/ANSI Z1.9, at inspection level II and the Acceptance Quality Limit (AQL) specified (see 6.2). Unless otherwise specified (see 6.2), modules tested shall represent largest (in volume) production sizes.

4.4.4 Selection of individual specimens. From each sample module selected in 4.4.3, one specimen shall be taken for each of the destructive tests specified in 4.5.3 (see footnote 1/ of table II for properties which can be determined with an individual specimen). The total number of specimens for each test shall contain an approximately equal number representing the top, middle, and bottom of the sample modules. For examples, specimens shall be selected from the top of one module, the middle of the second module, the bottom of the third module, etc., to obtain a representative sample.

4.4.5 Lot acceptance or rejection. Lots shall be accepted or rejected for each test separately and in accordance with ASQ/ANSI Z1.9, at acceptance level II and the Acceptance Quality Limit (AQL) specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:42Z
Check the source to verify that this is the current version before use.

4.5 Conformance inspection and tests.

4.5.1 Lot. For the purpose of sampling, examination, and test, a lot shall consist of one batch of microspheres and resin mixed at the same time in the same mixer and subsequently poured into molds to form on or more modules.

4.5.2 Examination. Each module shall be examined for size and form, dimensional tolerance, color, and workmanship to determine conformance with the requirements of this specification. Modules found not to be in conformance with the requirements of this specification shall be cause for rejection.

4.5.3 Conformance tests. Conformance tests shall consist of the tests specified in 4.6 except that 4.6.7.1 shall not apply. Physical properties of test specimens shall meet all the requirements of 3.2 and table II. The tests of 4.6.9 and 4.6.10 shall be run only on the first, middle, and last, or as specified (see 6.2).

4.6 Test methods. Each individual test value shall be reported.

4.6.1 Selection of test specimens. Test specimens shall be selected as specified in 4.4.1 for first article tests and 4.4.2 for conformance tests.

4.6.2 Standard test specimen. Unless otherwise specified (see 6.2), one standard 1-inch diameter by 2 inches long dimensional tolerance shall be plus or minus 0.1 inch.

4.6.3 Conditioning of test specimens. Test specimens shall be conditioned in a standard laboratory atmosphere of 73.4±2 °F (23±1.1 °C) and 50±2 percent relative humidity for a minimum of 24 hours prior to testing.

4.6.4 Test conditions. Tests shall be conducted in a standard laboratory atmosphere of 73.4±2 °F (23±1.1 °C) and at 50±2 percent relative humidity unless otherwise specified in the testing method or in this specification.

4.6.5 Tests for density.

4.6.5.1 With resin skin. Prior to sectioning the sample, the overall density of the sample with the skin (see 4.2.1) shall be determined. Measure the sample to the nearest 0.1 inch and calculate the volume in cubic inches, and weigh to the nearest 1.0 gram. Calculate density by:

$$\frac{\text{Weight of sample (grams)}}{\text{Volume of sample (cu. in.)}} \times 3.81 = \text{Density (lbs./cu.ft.)}$$

4.6.5.2 Density of specimens (without resin skin). One-inch diameter by 2-inch long cylindrical test specimens shall be used to determine the density of the syntactic material in accordance with ASTM D1622.

4.6.6 Exposure tests. Before physical tests are performed, the appropriate specimens shall be exposed continuously in a liquid for one week at 4,500 psig hydrostatic pressure for type I and at 10,000 psig hydrostatic pressure for type II. Unless otherwise specified (see 6.2), the exposure liquid for immersion tests shall be tap water.

4.6.7 Liquid absorption. Expose 1-inch diameter by 2-inch long tests specimens to test liquid at 75±10 °F for 10 seconds (see 4.6.6). Wipe specimen dry in a paper towel and immediately weigh to the nearest milligram. Expose specimens for one week in the test liquid at 4,500 psig hydrostatic pressure for type I and at 10,000 psig hydrostatic pressure for type II. Remove specimens from the test fluid, wipe dry in a paper towel and immediately weigh to the nearest milligram. Calculate liquid absorption by:

$$\frac{\text{Final weight} - \text{initial weight}}{\text{Initial weight}} \times 100 = \text{percent liquid absorption by weight}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:42Z
Check the source to verify that this is the current version before use.

4.6.7.1 1,000-cycle fatigue test (for first article only). Percent water absorption after exposure to a 1,000-cycle fatigue test shall be obtained after exposure to the following fatigue cycle:

a. Expose 1-inch diameter by 2-inches long test specimens to test liquid at 75±10 °F for 10 seconds. Wipe specimen dry in a paper towel and immediately weigh to the nearest milligram. Expose specimens for 1,000 cycles in the test liquid in accordance with the following test cycle:

(1) 4-minute rise in hydrostatic pressure:

(a) From atmospheric to 4,500 psig for type I only.

(b) From atmospheric to 10,000 psig for type II only.

(2) Hold pressure at:

(a) 4,500 psig for 12 minutes for type I only.

(b) 10,000 psig for 12 minutes for type II only.

(3) Release pressure at a rate of approximately 10 seconds/1000 psi so that:

(a) Pressure is reduced to atmospheric in 45±10 seconds for type I material only.

(b) Pressure is reduced to atmospheric in 100±20 seconds for type II material only.

(4) Hold 4 minutes at atmospheric pressure.

(5) Repeat for 1,000 cycles.

b. Remove specimens from the test fluid after completion of the 1,000-cycle tests. Wipe dry in a paper towel and immediately weigh to the nearest milligram. Calculate water absorption in accordance with formula in 4.6.7.

4.6.8 Uniaxial compressive tests. One-inch diameter by 2-inch long test specimens shall be tested in accordance with ASTM D695 to determine the compressive strength, 0.2 percent offset compressive stress, and the compressive modulus.

4.6.9 Tensile strength. The tensile strength test shall be in accordance with ASTM D638 using one-half inch thick, type II specimens as described therein (see 6.4).

4.6.10 Shear strength. The shear strength shall be determined in accordance with ASTM D732.

4.6.11 Bulk modulus. This method describes the procedure for determining the bulk modulus of a syntactic buoyancy material by means of a piston volume displacement method.

4.6.11.1 Apparatus. The following apparatus shall be used.

a. Cylinder and piston (see 6.5).

b. Hydrostatic pressing chamber.

c. I.D. = 2¼ inches.

d. O.D. = 4½ inches.

e. Working height = 10 inches.

f. Pressure rating = 15,000 psig.

g. Tonnage required = 29.8T.

h. Chamber material – UNS G43400 (AISI 4340) Heat Treated to 140,000 psi., Yield Strength. Source for alloy is ASTM DS56L-EB or SAE J1086.

i. Piston seal: Buna N “O” ring with PolyTetraFluoroEthylene (PTFE) backups.

4.6.11.2 Test specimen. The size of the test specimen shall be 4 by 1½ by 1½ inches (148 ml.).

4.6.11.3 Procedure. The procedure shall be as follows:

a. The high-grade alloy steel cylinder is partially filled with 500 ml. of water (fresh distilled water), and a stainless-steel block measuring 4 by 1½ by 1½ inches (148 ml) is inserted into the chamber.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:42Z
Check the source to verify that this is the current version before use.

b. The water in the cylinder is then compressed by the piston to 4,500 psig on a compression machine for type I and to 10,000 psig for type II using a rate of cross-head travel of 0.05 in./min.

c. The piston displacement and hence the change in volume of the water is recorded automatically on the machine chart. (a dial gage may be used in lieu of a recorder).

d. The steel block is removed and the syntactic specimen is introduced into the cylinder with the same volume of water present.

e. The combination of water and syntactic specimen is also subjected to 4,500 psig hydrostatic pressure on the compression machine for type I and to 10,000 psig for type II using a rate of cross-head travel of 0.05 in./min.

4.6.11.4 Calculations. Calculations shall be made as follows:

a. The volume change of the syntactic specimen is then determined by subtracting the compression of the water from the compression of the combination.

b. A correction is made for water absorption by weighing the specimen before and after exposure. (the weight gain is converted to volume and subtracted from the actual ΔV_s below.)

c. The relationship between the volume change of water and volume change of the syntactic material may be expressed in the following way:

$$k = -\frac{\Delta V_s}{V_s} = -(\frac{\Delta V_c - \Delta V_W}{V_s})$$

Where:

k = compression.

ΔP = change in pressure.

ΔV_s = Change in volume of syntactic material (corrected for absorbed water).

ΔV_c = Change in volume of combination, syntactic material and water.

ΔV_W = Change in volume of water.

V_s = Original volume of syntactic material.

Where: Bulk modulus is $E_B = \frac{-\Delta P}{\Delta V_s} - V_s$

4.6.12 Hydrostatic compressive strength. The hydrostatic compressive strength test shall be determined as specified in Appendix A.

4.7 Inspection for preparation for delivery. DELETED.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.1.1 Syntactic buoyancy material. DELETED.

5.1.1.1 Packaging. DELETED.

5.1.1.2 Packing. DELETED.

5.1.1.3 Marking. DELETED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:42Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Syntactic buoyancy material is intended for use in Navy submerged research vehicles, underwater floats, encapsulants, etc. As such, it will be directly exposed to the sea or some other liquid medium which provides hydrostatic forces on the exposed surfaces.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Size, form, and number of modules required (see 3.5).
d. Dimensions and dimensional tolerances required (see 3.6).
e. Requirements for resin skin (see 3.8).
f. If first article inspection is other than specified (see 4.3).
g. If representation of modules is other than specified (see 4.4.3).
h. AQL of 4.0 (see 4.4.3 and 4.4.5).
i. Frequency of tensile and shear tests, if other than specified (see 4.5.3).
j. If standard test specimen is other than specified (see 4.6.2).
k. When exposure liquid other than tap water is required (see 4.6.6 and 4.6.7).
l. Packaging requirements (see 5.1).

6.3 Definition. Syntactic buoyancy material (or syntactic foam, as known in the trade) is defined as a material consisting of hollow sphere fillers incorporated in a resin matrix.

6.4 Tensile strength test specimen. Type I specimens as described in ASTM D638 have been found unsuitable for tensile strength tests because syntactic material is brittle, and type I specimens have occasionally broken in the jaws.

6.5 Cylinder and piston apparatus. Apparatus may be procured from Loomis Products Company, Levittown, PA, or equal.

6.6 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.6. The contracting officer should also include specific instruction in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.7 Technical authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.8 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:42Z
Check the source to verify that this is the current version before use.

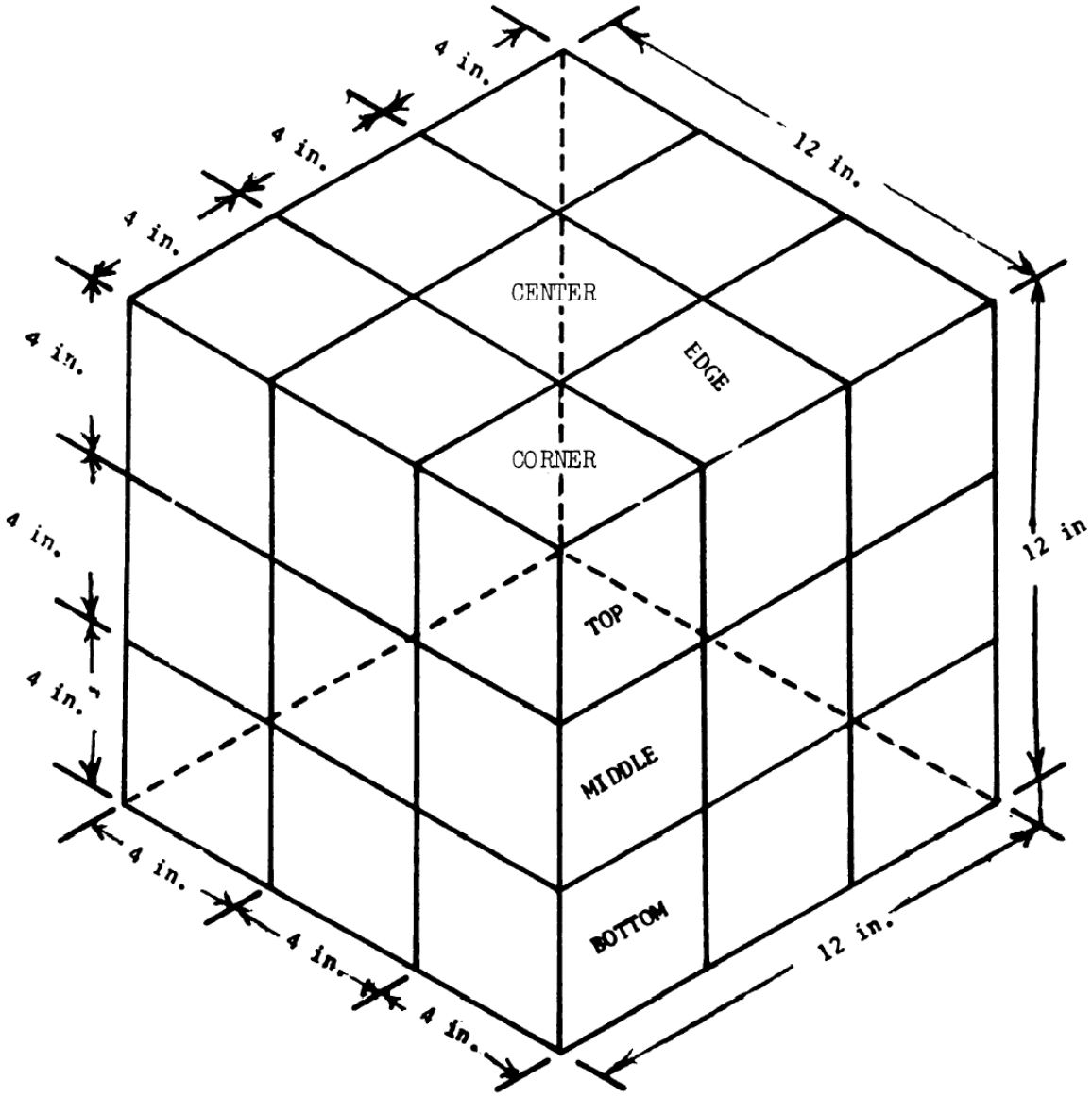


FIGURE 1. Stratification of first article sample.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:42Z
Check the source to verify that this is the current version before use.

A.1 SCOPE

A.1.1 This test procedure is designed to quickly and accurately determine the hydrostatic compressive strength of any syntactic foam capable of withstanding 1000-psi or higher hydrostatic loads. The test is simple, yet accurate to within 100 psi. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 SUMMARY OF METHOD

A.2.1 This test method consists of applying a hydrostatic compressive load to a syntactic foam specimen via a test machine cross-head which bears down on a sliding piston contained in a high-pressure cylinder. The load required to cause failure of the specimen shall be taken from a load-deflection plot of the applied load as observed on the test machine dial.

A.3 SIGNIFICANCE

A.3.1 This test provides information about the usefulness of a syntactic foam for a given hydrostatic pressure application. The test is limited to foams with a hydrostatic compressive strength of more than 1000 psi because the accuracy of determinations below this level are questionable.

A.4 INTERFERENCES

A.4.1 Some of the problems which can be encountered in this method and which should be avoided are:

a. Entrapped air in the test vessel.

b. Leaks in the hydraulic system.

c. Friction between the seal and cylinder walls.

A.4.2 The test procedure is to simply place the test specimen of standard size into the test vessel of standard size containing a standard volume of water, bleed off the entrapped air and apply a load at a standardized rate of speed (cross-head speed) with a test machine and plot a load-deflection curve.

A.5 APPARATUS

A.5.1 Any suitable testing machine capable of control of constant rate-of-crosshead movement and consisting of a drive mechanism, a load indicator and an extensometer shall be used. A test machine such as specified in method ASTM D695, Test for Compressive Properties, can be used.

A.5.2 Piston-cylinder apparatus. A test vessel consisting of a piston-cylinder apparatus, size 1.5 inches inside diameter by 6 inches deep (working dimensions) such as illustrated in figure 1A may be used. The apparatus shall be rated for 35,000 psi minimum. (NOTE: A 1.5-inch inside diameter by 6-inch deep [working-dimensions] test vessel rated for 35,000 psi supplied by Loomis Engineering and Manufacturing Co., Caldwell, N.J., or equivalent, shall be used).

A.6 SAFETY PRECAUTIONS

A.6.1 The usual safety precautions applicable to systems under high hydraulic loads shall be observed.

A.7 Sample Size

A.7.1 The sample shall consist of a 1±0.1-inch diameter cylinder, 2±0.1 inches long.

A.8 PROCEDURE

A.8.1 The hydrostatic compressive specimen shall be placed in the isostatic pressing cylinder partially filled with 210 ml. of water.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:42Z
Check the source to verify that this is the current version before use.

A.8.2 The piston shall be placed in the cylinder and the entrapped air and 10 ml. of water shall be bled off from the vent hole, and the air vent shall then by tightly closed.

A.8.3 The piston cylinder shall be loaded on a compression machine using a rate of cross-head travel of 0.20 in./min. (Preliminary investigations indicate that hydrostatic compressive strength is not significantly affected by load rate.)

A.8.4 The hydrostatic compressive load shall be that point at which a substantial drop in load is evidenced on the compression machine's dial gage or the load-deflection curve.

A.8.5 The load shall be released upon reaching the hydrostatic compression load and the equipment shall be prepared for the next test.

A.9 INTERPRETATION OF RESULTS

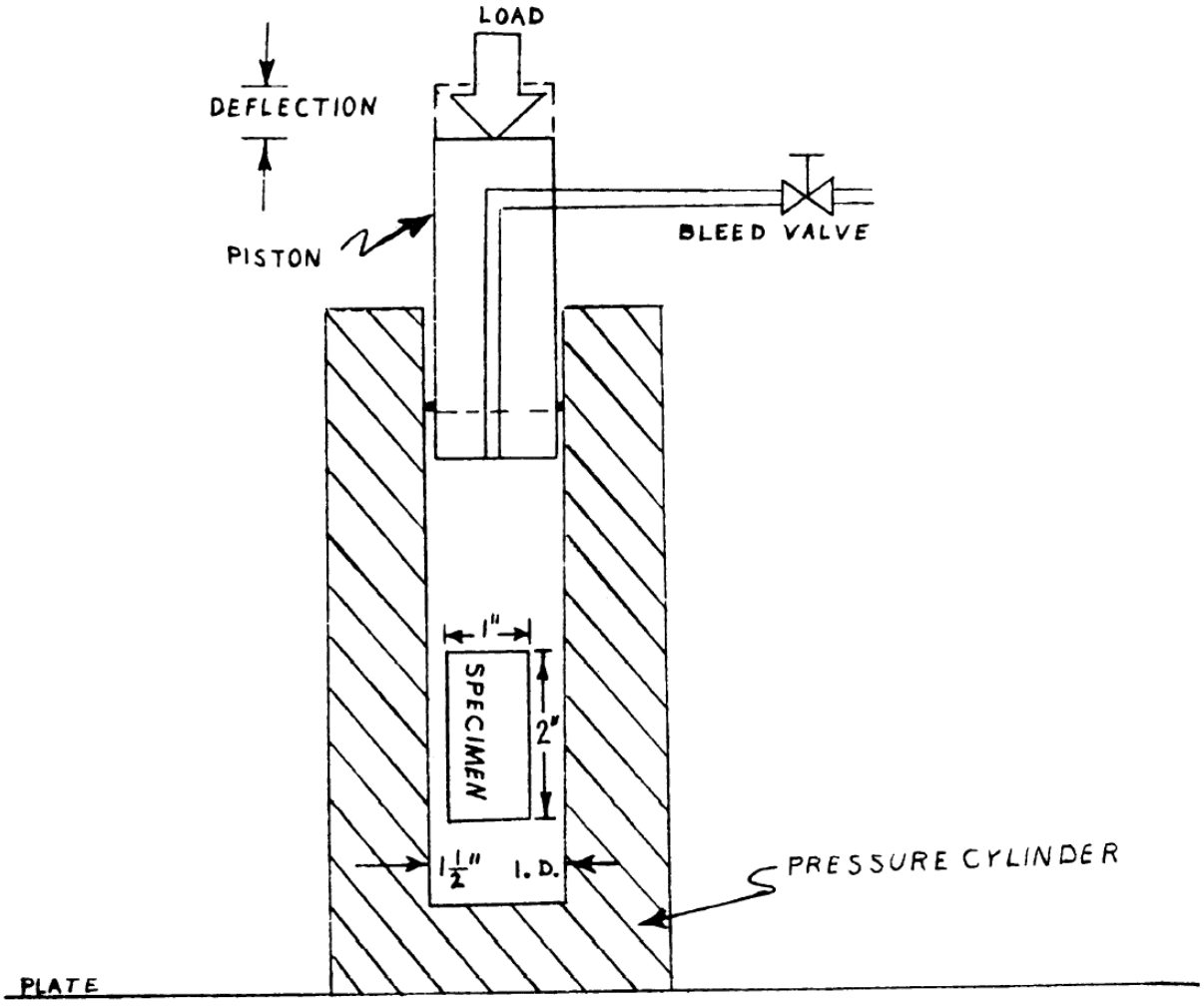
A.9.1 In the case of high strength foams, the failure point is quite obvious, manifested by a sharp break in the load-deflection curve (or a sudden drop in applied load on the test machine dial). Lower strength foams may fail by gradual compression or water absorption and this mechanism of failure is not immediately obvious. This is where the load-deflection curve shall be employed to determine actual failure by determining what percent offset yield corresponds to the arbitrarily chosen water absorption or specimen deformation corresponding to specimen failure.

A.10 CALCULATIONS

A.10.1 Ultimate hydrostatic crush strength shall be calculated (in psi) from the applied hydrostatic load (indicated by the test machine dial or the arbitrarily chosen yield point of the load-deflection curve) divided by the cross-sectional area of the test vessel's piston:

$$\text{Ultimate hydrostatic strength} = \frac{\text{Load (in pounds)}}{\text{Cross-sectional area of test vessel piston}}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:42Z
Check the source to verify that this is the current version before use.



FIGUR E 1A. Schematic of hydrostatic compressive test equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:42Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
DLA – GS

Preparing activity:
Navy – SH
(Project 9330–2021–002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:42Z
Check the source to verify that this is the current version before use.